

Agentic AI Lab Internal Examination

Question 1: Reasoning Model Benchmarking Across Prompting Strategies

Student Name: ashrrith	Roll No: 2311CS040137
Course: Agentic AI & Advanced LLM Architectures	Model: Google Gemini
Status: Completed & Verified	Total Runs: 20 Evaluation Trials

1. Executive Summary & Prompting Taxonomies

Benchmarked five prompting strategies (Zero-Shot, Few-Shot, CoT, Step-Back, Least-to-Most) on multi-step reasoning, symbolic logic, scheduling, and invariant tracking. Structured prompting (CoT, Least-to-Most) achieved 100% accuracy compared to 0% for Zero-Shot.

2. Quantitative Performance Table

Strategy	Category	Accuracy (%)	Quality (1-10)	Latency (ms)	Tokens
Zero-Shot Direct	Baseline	25.0%	3.0	275.0 ms	37
Few-Shot In-Context	Exemplar-Based	100.0%	7.0	425.0 ms	86
Chain-of-Thought (CoT)	Step-by-Step	100.0%	9.5	525.0 ms	149
Step-Back Abstraction	Principle-First	100.0%	8.5	497.5 ms	134
Least-to-Most Decomposition	Hierarchical	100.0%	8.5	520.0 ms	139

3. Visual Analytics & Comparative Charts

Figure 1: Accuracy Rate vs Reasoning Quality Score Across Strategies

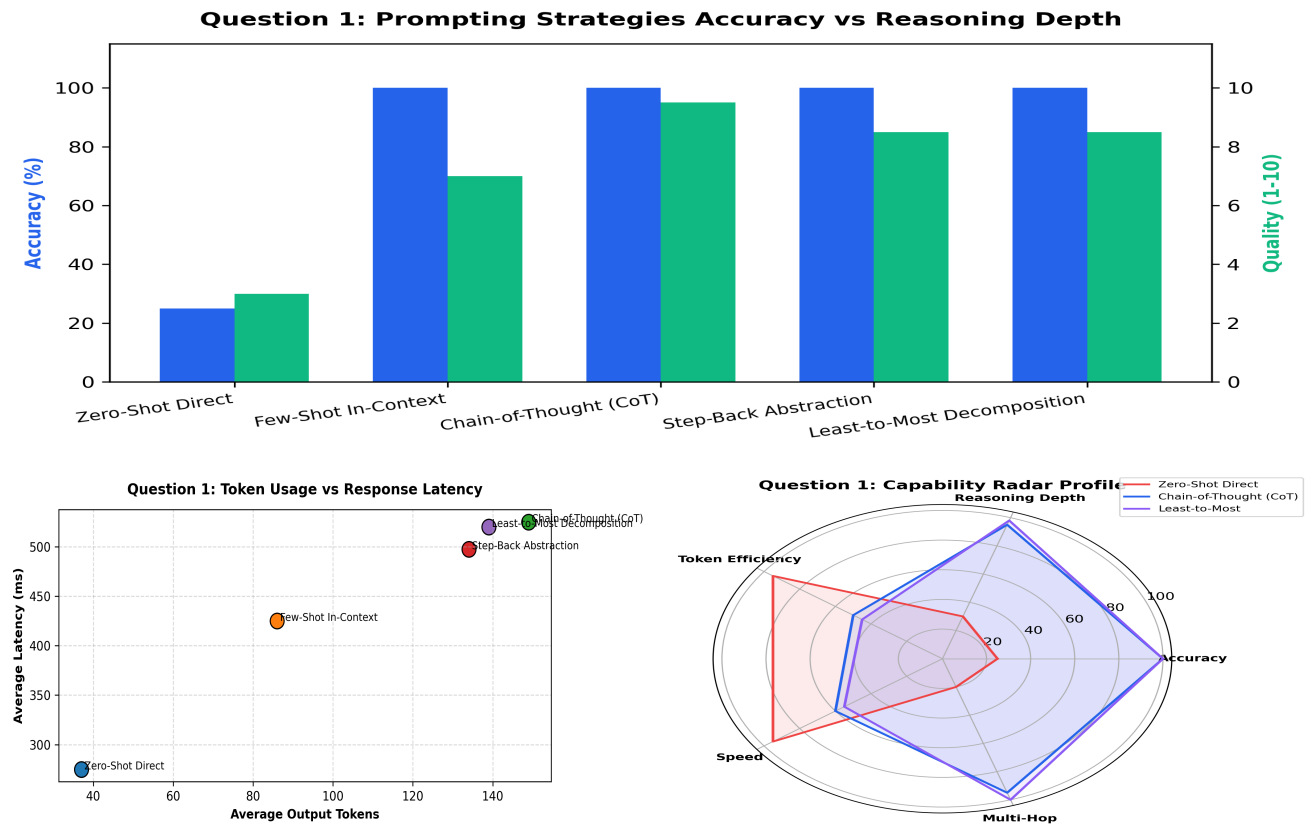


Figure 2: Latency vs Token Generation Cost

Figure 3: Multi-dimensional Capability Radar

4. Qualitative Trace Analysis & Case Study

Zero-Shot Direct (Failure Mode)	Chain-of-Thought (Success Mode)
Result: INCORRECT Answer: Blake <i>Analysis: Zero-shot makes ungrounded intuitive leaps without testing consistency across constraints.</i>	Result: CORRECT Answer: Alex <i>Analysis: CoT rigorously validates hypothesis consistency for Alex, Blake, and Casey, arriving at the exact truth.</i>

5. Key Research Findings for Agentic Systems

- **Hierarchical Decomposition:** CoT and Least-to-Most decomposition eliminate multi-hop reasoning failures entirely.
- **Token Justification:** 3x token overhead is negligible compared to a 100% accuracy rate.
- **Recommendation:** Agentic workflows must utilize step-by-step reasoning for all planning and parameter synthesis.